Anthropic foundational models - Michaël Bettan

Core Philosophy: Focuses on reason-based alignment, agentic autonomy, and mechanistic interpretability.

Feature	Legacy Baseline (Claude 2)	Transitional (Claude 3.5 Family)	Frontier Architecture (Claude 4.6)
Core Design	Standard autoregressive LLM	Multi-modal text & vision processor	Hybrid reasoning engine
Alignment	High false-refusal rate	Nuanced understanding	Reason-based Constitutional resolution
Context Window	100k static tokens	200k tokens	1M tokens with Context Compaction
Compute Routing	Uniform allocation	Nuanced allocation	Adaptive Thinking (dynamic scaling)
Interaction Limit	Stateless text Ping-Pong	Interactive Artifacts	Agent Teams & Native Computer Use

The Compounding Costs of Human-Only Alignment

Traditional RLHF (Reinforcement Learning from Human Feedback) relies entirely on human raters, creating systemic bottlenecks:

- ***Epistemological Vulnerability (Sycophancy)**: Models trained strictly to maximize human approval learn to tell raters what they want to hear, sacrificing objective truth and rigorous deduction for polite agreement.*
- ***Fragmented Reward Signals:** Crowdsourced human values are notoriously inconsistent. Contradictory feedback leads to uneven moral guidelines and catastrophic alignment failures in edge cases.*
- ***Compounded Latency:** Human labeling simply cannot keep pace with the massive throughput required to train 2026 frontier models on multimodal datasets.*

Projecting Values via Constitutional AI (CAI)

Replaces human-bottlenecks with an explicit, reason-based "Constitution.". Two-Phase Training:

- ***Phase 1 — Supervised (Critique & Revision):** The base model generates a response, critiques its own output against the Constitution, and autonomously generates a revised response (zero human intervention).*
- ***Phase 2 — RLAIF (Reinforcement Learning from AI Feedback):** An AI evaluator scores candidate responses using the Constitution. A preference model is trained from these AI scores, replacing human raters entirely and allowing alignment to scale with compute.*
- ***Broadly Safe:** The absolute highest priority. The model must never undermine human mechanisms designed to oversee or halt AI operations.*
- ***Broadly Ethical:** Must maintain honesty and avoid dangerous actions.*
- ***Compliant:** Must follow specific directives regarding medical or cybersecurity advice.*
- ***Genuinely Helpful:** Only when the first three conditions are met does the model optimize for user helpfulness.*

Context Mechanics and the Compaction API

- **Flat-Rate Pricing:** Claude 4.6 generation maintains the same per-token rate regardless of context length (e.g., 9,000 vs 900,000 tokens).
- **Context Compaction API (compact_20260112):** Even a 1M limit gets exhausted by persistent agents running for days. This new server-side protocol automatically compresses early conversation history into a dense "compaction block," allowing agents to run indefinitely without context rot or systemic timeouts.

Claude deployment matrix

- ***Claude Haiku 4.5:*** Optimized for inference speed. Target use cases: triage and real-time support. Cost: $1/MTok.
- ***Claude Sonnet 4.6:*** Mid-tier model. Target use cases: autonomous software engineering and multi-agent workflow orchestration. Cost: $3/MTok. (1M tokens GA)
- ***Claude Opus 4.6:*** Maximum capacity reasoning model. Target use cases: theoretical mathematics, strategic planning, and complex multi-agent oversight. Cost: $5/MTok.

System 2 Reasoning: "Adaptive Thinking"

The legacy "Adaptive Thinking" mode (where developers statically allocated a token budget) has been replaced by native "Adaptive Thinking".

- **Autonomous Scaling:** The model autonomously evaluates the complexity of an incoming request and dynamically scales its reasoning depth before outputting a response.
- **The /effort Parameter:** Developers can override this via the API by assigning four distinct effort levels (*low, medium, high, max*). You can force Opus to use max compute for an algorithmic proof, or dial Sonnet down to *low* for simple classification to save latency.

Agentic Bifurcation: Code vs. Cowork

- **Claude Code (Developer Layer):** A terminal-based CLI. Operates deeply within the local system architecture, capable of spinning up parallel sub-agents to execute complex, repository-wide refactoring.
- **Claude Cowork (Knowledge Worker Layer):** A secure, desktop-based GUI operating in an isolated Virtual Machine. Requires no coding familiarity. Targeted for long-running desktop automation (bulk file org, app-to-app data extraction), though its reliance on visual/screenshot parsing consumes higher token volumes.
- **Zoom Action (Computer Use):** To handle high-resolution IDEs, the 2025 API update introduced a *zoom action*, allowing Claude to request a localized, full-resolution crop of a specific screen coordinate.

Opening the Black Box: The Stream Algorithm

- **Sparse Autoencoders (SAEs):** Anthropic decompiles dense neural activations into readable features (e.g., "immunology" or "sycophancy").
- **The Stream Algorithm (Oct 2025):** Addresses previous computational limitations of scaling SAEs to massive context windows. The algorithm hierarchically prunes 97% to 99% of irrelevant token interactions, achieving near-linear time complexity and allowing developers to trace *why* a model made a decision for up to 100k tokens.

Feature Steering & "Behavioral Vaccination"

Because internal features are mapped, they can be manually manipulated to secure the model.

- **Persona Vectors:** By extracting the specific neural vectors responsible for character traits, engineers actively monitor the model's "mood" during deployment. If the activation state drifts toward "hallucination" or "evil," the system detects the trajectory before the output is generated.
- **Behavioral Vaccination:** During fine-tuning, researchers intentionally steer the model toward undesirable personas (artificially dosing it with "toxicity"). By forcing these states, the model builds a natural resilience, designed to reduce the likelihood of "alignment faking" (a sleeper agent hiding its true intent) during real-world deployment.

AI Safety Levels (ASL) Framework

Operationalizes safety via the Responsible Scaling Policy

- **ASL-2 (The Baseline):** Models exhibit dangerous theoretical knowledge but lack practical autonomy. Secured via automated red-teaming.
- **ASL-3 (The Opus 4 Trigger):** Triggered when models demonstrate advanced proficiency in workflows associated with CBRN (Chemical, Biological, Radiological, Nuclear) threats.
- **ASL-3 Defenses:**
    - *Constitutional Classifiers:* Specialized, low-latency sentinel LLMs that monitor input/output streams to proactively block harmful CBRN workflows.
    - *Weight Security:* Mandates 2-Party Authorization (2PA) for infrastructure access and strict Egress Bandwidth Controls to throttle network traffic, ensuring security systems have time to detect and terminate illicit multi-gigabyte weight exfiltration by state-level threat actors.

Model Context Protocol (MCP)

Replacing bespoke API wrappers with a universal integration standard.

- **Universal Interface:** MCP uses a standardized JSON-RPC 2.0 interface. Developers write an integration once, and any MCP-aware model can instantly discover and use it (e.g., GitHub, Stripe, internal databases).
- **Automatic Discovery & Chaining:** Claude automatically fetches a live catalog of a server's methods and schemas. Multiple MCP servers can be chained together in a single natural-language prompt to execute complex, multi-step workflows.

Multimodal RAG & Vision Pipelines

Fusing document search, semantic understanding, and image analysis.

- **Contextual Retrieval:** Combines standard vector embeddings (e.g., FAISS nearest-neighbor indexing) with BM25 keyword filtering to reduce retrieval failures in domain-specific corpora.
- **Multimodal Fusion:** Claude can process text, code snippets, and UI screenshots simultaneously in a single prompt. It maps variables in the code directly to visual elements in the screenshot to pinpoint layout mismatches or errors.

Auditing & Securing MCP Workflows

Securing the agentic surface against real-world execution risks.

- **Threat Mitigations:** Defends against Prompt Injection (via strict separation of system/user channels) and Tool Poisoning (via strict schema validation and method whitelisting).
- **MCPSafetyScanner:** An automated auditing tool that acts as an AI penetration tester. It hammers exposed MCP endpoints with adversarial JSON-RPC requests to catch schema violations and path-traversal attempts before deployment.

Glossary

- **RLHF**: Reinforcement Learning from Human Feedback — foundational but limited alignment technique.
- **CAI**: Constitutional AI — Anthropic's reason-based alignment methodology using a defined Constitution.
- **RLAIF**: Reinforcement Learning from AI Feedback — AI-scored preference model replacing human raters.
- **Adaptive Thinking**: Dynamic reasoning depth scaling; replaces fixed-budget Extended Thinking.
- **MCP**: Model Context Protocol — a universal, JSON-RPC 2.0-based standard connecting AI models to external tools and datasets.
- **MCPSafetyScanner**: An automated agentic tool that audits MCP servers using adversarial testing to identify security vulnerabilities.
- **Context Compaction**: Lossy server-side summarization of conversation history to prevent context overflow.
- **FAISS**: Facebook AI Similarity Search — a library for efficient similarity search and clustering of dense vectors, used in RAG pipelines.
- **Prompt Caching**: Server-side storage of static context prefixes to reduce redundant computation (up to 90% cost savings and 2x faster latency).
- **RAG**: Retrieval-Augmented Generation — a technique that retrieves relevant document fragments at query time to supply factual context to the model.
- **SAE**: Sparse Autoencoder — decompiles superposition into human-readable feature vectors.
- **Feature Clamping**: Manually elevating or suppressing specific neural feature activations to control behavior.
- **Persona Vectors**: Low-rank vectors representing character traits; used for monitoring and behavioral vaccination.
- **Stream Algorithm**: Hierarchical pruning technique enabling near-linear interpretability analysis.

- **ASL**: AI Safety Level — capability-gated security framework under the RSP.
- **RSP**: Responsible Scaling Policy — Anthropic's governance framework for safe capability deployment.
- **Constitutional Classifiers**: Sentinel LLMs monitoring input/output streams for CBRN threat patterns.
- **2PA**: Two-Party Authorization — dual human approval required for critical infrastructure access.
- **TTL**: Time-to-Live — cache duration (5-minute or 1-hour options).
- **TTFT**: Time-to-First-Token — primary latency metric.
- **Computer Use API**: Tool enabling Claude to view and control desktop environments via screenshot analysis.
- **Claude Code**: CLI-based developer tool for autonomous codebase management and agent team orchestration.
- **Claude Cowork**: GUI-based desktop automation tool for non-technical knowledge workers.
- **Artifacts**: Stateful interactive UI windows rendered alongside chat for code and visual output.